

ISPilot

Status: ☒ API Deployed to Cloud Run (Production)

API Endpoint: <https://ispilot-api-46y2f3tyja-uc.a.run.app>

ISPilot is an enterprise AI platform designed to provide retail business users with natural language access to Intelligent Shelf operational data.

The platform transforms operational metrics into actionable business insights through a specialized multi-agent architecture built on:

- Google ADK
- Vertex AI Agent Engine
- Gemini
- BigQuery
- Secret Manager

Instead of navigating dashboards, reports, or SQL queries, users can interact directly with retail data using natural language.

Business Vision

ISPilot is not a chatbot.

ISPilot is a business intelligence platform powered by AI agents.

The objective is to create reusable business capabilities that can be consumed through:

- AI Agents
- APIs
- Microsoft Teams
- Copilot Studio
- Future Enterprise Applications

Business users should be able to ask:

How is Talca Colin performing?

How can Talca Colin improve?

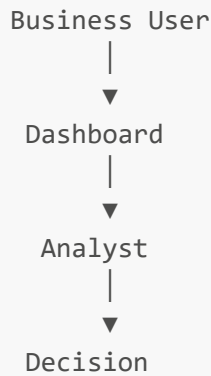
What are today's priorities?

Which stores need immediate attention?

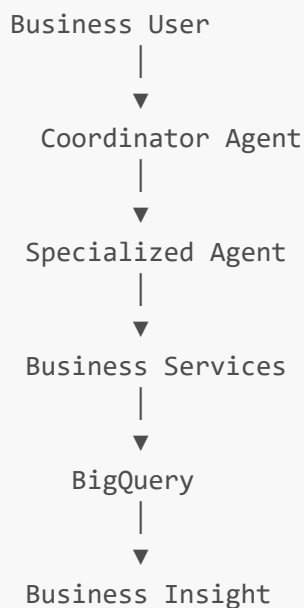
without requiring dashboards, reports or analyst intervention.

What Makes ISPilot Different

Traditional Model



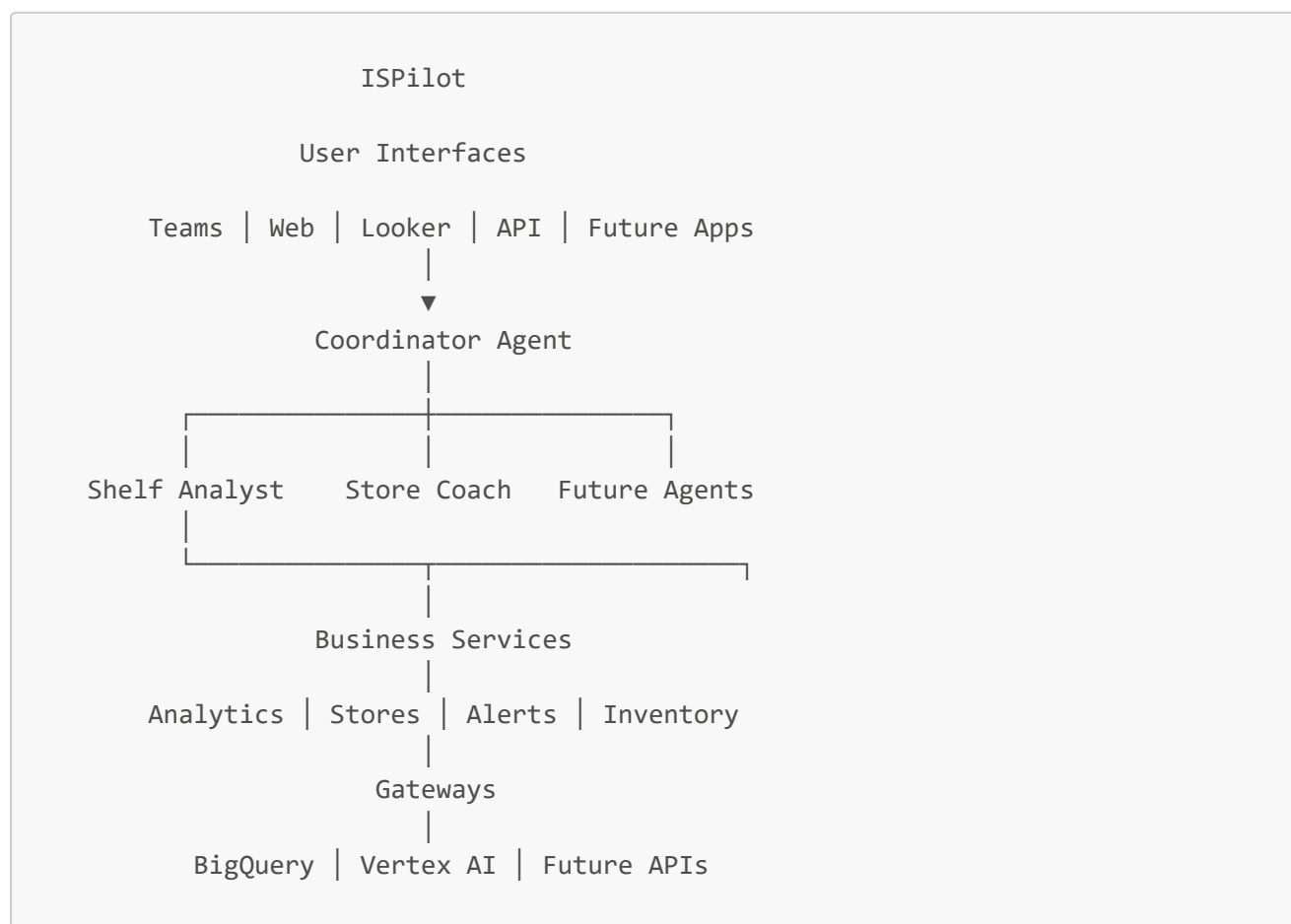
ISPilot Model



Benefits:

- Faster decision making
- Reduced analyst dependency
- Consistent recommendations
- Natural language interaction
- Reusable business services

Current Architecture



Multi-Agent Architecture

ISPilot uses a hierarchical multi-agent architecture implemented with Google ADK.

The Coordinator Agent acts as the single entry point and delegates requests to specialized business agents.

Agent Responsibilities

Agent	Business Question
Coordinator	Who should answer?
Shelf Analyst	What is happening?
Store Coach	What should I do?
Root Cause Agent	Why is it happening? <i>(future)</i>
Executive Agent	What does it mean for the business? <i>(future)</i>

Coordinator Agent

Responsibilities:

- Intent detection
- Agent routing
- Delegation
- Conversation orchestration

Coordinator does **not**:

- Query BigQuery
- Execute business logic
- Generate recommendations

Its responsibility is orchestration.

Shelf Analyst

Mission:

What is happening?

Examples:

How is Chile today?

How is Talca Colin performing?

What are the rankings?

What are the trends?

Compare today versus yesterday.

Current Capabilities:

- Country Summary
- Store Summary
- Store Name Resolution
- Store Code Resolution
- Ambiguous Store Detection
- Rankings
- Daily Comparisons
- Historical Trends
- Natural Language Analytics

Example Flow:

```
graph TD
    User --> Coordinator
    Coordinator --> Shelf_Analyst[Shelf Analyst]
    Shelf_Analyst --> Business_Services[Business Services]
    Business_Services --> BigQuery
```

Store Coach

Mission:

What should I do?

Examples:

How can Talca Colin improve?

What should the manager do?

What actions should be prioritized?

What should be fixed first?

Current Capabilities:

- Operational Recommendations
- Store Improvement Guidance
- Corrective Actions
- Prioritization Support
- Action Plans

Store Coach consumes analytical outputs and converts them into actionable recommendations.

Project Structure

intelligent-shelf-agents/

```
├── agents/
│   ├── coordinator/
│   │   ├── agent.py
│   │   ├── requirements.txt
│   │   ├── .agent_engine_config.json
│   │   └── subagents/
│   │       ├── shelf_analyst/
│   │       ├── store_coach/
│   │       ├── executive/
│   │       └── root_cause/
│
│   ├── common/
│   │   ├── config.py
│   │   ├── bigquery_client.py
│   │   ├── gateway_gemini.py
│   │   └── gateway_token.py
│
│   ├── gateways/
│   │   ├── nsg_gateway.py
│   │   └── store_gateway.py
│
│   ├── services/
│   │   ├── analytics_service.py
│   │   └── store_service.py
│
│   ├── repositories/
│
│   ├── domain/
│
│   └── tests/
```

Business Terminology

Business Term	Technical Metric
SNSG	N_ON_SHELF
En Góndola	ON_SHELF
Bodega	OOS_SHELF
Quiebre	OOS_STORE

API Integration

The ISPilot agents are exposed through a REST API running on Google Cloud Run.

Production API

Endpoint: <https://ispilot-api-46y2f3tyja-uc.a.run.app>

Authentication: OAuth 2.0 Bearer Token (Workload Identity)

Status: ☒ Active and responding

Deployment Details:

- Service Account: sa-tot-osa@corp-stro-salesinventory-prod.iam.gserviceaccount.com
- Region: [us-central1](#)
- Memory: 1Gi
- CPU: 2
- Authentication: Workload Identity (no credential files)
- Session Backend: Cloud Firestore (with in-memory fallback)

Quick API Test

```
# Get authentication token
AUTH_TOKEN=$(gcloud auth print-identity-token --audiences https://ispilot-api-46y2f3tyja-uc.a.run.app)

# Test the API
curl -X POST https://ispilot-api-46y2f3tyja-uc.a.run.app/chat \
  -H "Content-Type: application/json" \
  -H "Authorization: Bearer $AUTH_TOKEN" \
  -d '{
    "user_id": "sebastian",
    "message": "How is Talca Colin performing?"
  }'
```

API Documentation

See [ispilot-api/README.md](#) for complete API documentation including:

- All endpoints (health, chat)
- Request/response schemas
- Authentication methods
- Session management
- Error handling

Business users should interact using business terminology rather than technical metric names.

Current Features

Shelf Analyst

Current Capabilities

- Country KPIs
- Store KPIs
- Store Name Resolution
- Store Code Resolution
- Rankings
- Daily Comparisons
- Historical Trends
- Natural Language Analytics

Examples

How is Chile today?

How is store 101?

How is San Bernardo Plaza?

How is San Bernardo?

What are the worst performing stores?

Store Coach

Current Capabilities

- Recommendations
- Prioritization
- Operational Guidance
- Action Plans
- Store Coaching

Examples

How can store 101 improve?

What should I do to improve SNSG?

What are the main priorities for this store?

What actions should be taken first?

Technology Stack

Core Platform

- Google ADK
- Gemini 2.5 Flash
- Vertex AI Agent Engine
- BigQuery
- Secret Manager
- Python 3.12

Architectural Patterns

- Multi-Agent Architecture
- Clean Architecture
- Gateway Pattern
- Service Layer
- Domain Objects
- Coordinator Orchestration Pattern

Current Status

Operational

- ☒ Coordinator Agent
- ☒ Shelf Analyst
- ☒ Store Coach
- ☒ Vertex AI Agent Engine
- ☒ BigQuery Integration
- ☒ Secret Manager Integration
- ☒ Multi-Agent Routing
- ☒ REST API Validation
- ☒ Vertex Playground Validation

Next Milestones

- Root Cause Agent
- Executive Agent
- Recommendation Engine

- Microsoft Teams Integration
- Copilot Studio Integration
- Enterprise APIs
- Observability & Telemetry

Deployment

Production deployments use Google ADK Agent Engine.

Deploy Command

```
adk deploy agent_engine \  
  --project=corp-stro-salesinventory-prod \  
  --region=us-central1 \  
  --display_name=intelligent-shelf-coordinator \  
  --extra_packages common \  
  --extra_packages services \  
  --extra_packages gateways \  
  --extra_packages repositories \  
  --extra_packages domain \  
  agents/coordinator
```

Runtime Configuration

Runtime configuration is centralized in:

```
agents/coordinator/.agent_engine_config.json
```

Example:

```
{  
  "service_account": "sa-tot-osa@corp-stro-salesinventory-  
prod.iam.gserviceaccount.com",  
  "env_vars": {  
    "IS_BQ_PROJECT": "corp-stro-salesinventory-prod",  
    "IS_BQ_DATASET": "acc_tot_cp_is_prd",  
    "IS_NSG_REPORT_VIEW": "vw_nsg_report",  
    "IS_CLASSIFICATION_VIEW": "vw_smartnsg_classification_cl_v2"  
  }  
}
```

Subagents do not require their own deployment configuration.

Authentication

Runtime Service Account

```
sa-tot-osa@corp-stro-salesinventory-prod.iam.gserviceaccount.com
```

Required Permissions

Secret Manager

```
roles/secretmanager.secretAccessor
```

Required Secret:

```
genai-gateway-jwt-prod
```

Service Usage

```
roles/serviceusage.serviceUsageConsumer
```

BigQuery

```
roles/bigquery.jobUser
```

```
roles/bigquery.dataViewer
```

Local Development

Authentication:

```
export GOOGLE_APPLICATION_CREDENTIALS=/path/to/sa-tot-osa.json
```

Start locally:

```
adk web agents/coordinator
```

Production runtimes running in Vertex Agent Engine must use the runtime service account and should not rely on local JSON credentials.

Future Channels

ISPilot is designed to be consumed through:

- Vertex Playground
 - REST APIs
 - Microsoft Teams
 - Copilot Studio
 - Enterprise Applications
 - Future Mobile Experiences
-

Documentation

Additional documentation is available in:

```
docs/  
├─ ISPilot-Platform-Overview.md  
└─ ISPilot-Enterprise-Architecture-And-Vertex-Agent-Engine-Guide.md
```

Document Purpose

ISPilot-Platform-Overview.md

Executive overview of the platform including:

- Business vision
- Multi-agent architecture
- Current capabilities
- Technology stack
- Strategic roadmap

ISPilot-Enterprise-Architecture-And-Vertex-Agent-Engine-Guide.md

Technical reference covering:

- Business architecture
- Multi-agent design
- ADK implementation
- Vertex Agent Engine deployment
- Security model
- REST API integration
- Troubleshooting history
- Operational runbook
- Teams and Copilot Studio integration strategy

Strategic Roadmap

Phase 1 ☒

- Country KPIs
- Store KPIs
- Store Resolution

Phase 2 ☒

- Store Coach
- Recommendations
- Action Plans

Phase 3

- Root Cause Analysis Agent

Phase 4

- Executive Agent

Phase 5

- Recommendation Engine

Phase 6

- Microsoft Teams Integration
- Copilot Studio Integration

Phase 7

- Enterprise APIs
- Agent Evaluations
- Observability & Telemetry

Phase 8

- Enterprise AI Platform Expansion

Long-Term Vision

The goal is to evolve ISPilot into an enterprise AI platform capable of supporting:

- Retail Operations
- Category Management
- Supply Chain
- Commercial Execution
- Executive Decision-Making

through a collection of specialized AI agents sharing a common business foundation.

The platform should provide natural language interaction while maintaining a clean separation between:

- AI Orchestration
- Business Rules
- Analytics Services
- Data Access Layers
- Enterprise Integrations